Evaluating Retrieval-Augmented Generation Pipelines: A Non-Invasive, Modular Benchmarking Framework for Automated RAG Configuration Analysis

Niclas Hart, Nicolas Weeger

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Abstract

Retrieval-Augmented Generation (RAG) has emerged as a dominant paradigm for grounding large language model outputs in factual, domain-specific knowledge. Yet practitioners face a vast, poorly understood configuration space spanning chunking strategies, embedding models, retrieval methods, and generation models. To overcome these limitations, we present a modular and extensible benchmarking framework designed to integrate seamlessly with existing RAG pipelines and automatically evaluate pipeline configurations. By covering the full RAG lifecycle and enabling systematic exploration of a large configuration space across all components the framework eliminates manual tuning while ensuring complete experiment reproducibility via MLflow. To maximize evaluation coverage, it combines LLM-based quality metrics with established numerical metrics from information retrieval and natural language generation.

The framework provides multiple pre-configured dataset adapters for evaluating RAG pipelines across diverse QA domains, while remaining extensible to custom domain-specific datasets. Validation is performed by benchmarking more than 100 configuration combinations on locally served open-weight models. The proposed approach enables systematic benchmarking of configurable RAG components, while being designed with modularity and extensibility in mind, aiming to simplify integration into existing RAG pipelines with minimal modifications.